A Guide to the Yellow Book

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1 Preface

The book *Conformal Field Theory* by Philippe Di Francesco, Pierre Mathieu, and David Sénéchal is, without doubt, an unavoidable entry point into conformal field theory. Owing to its bright yellow cover and its more than 800 pages, it is affectionately called "the Yellow Pages". This monumental work is renowned for its comprehensiveness.

However, as with most physics books, for many concepts the book adopts an approach of "understand through use": the concepts are shrouded in a thin mist—their rough outlines can be perceived, but their precise definitions often have to be worked out by the reader or pieced together from the context. For beginners, or for learners trying to pin down the details rigorously, this poses a considerable obstacle. (After all, even in Weinberg's quantum field theory one unavoidably encounters whispers from the old gods, such as "$|p\rangle$ transforms like a momentum".)

The purpose of these notes is precisely this. They are not a substitute for, or a criticism of, the Yellow Pages, but a complementary "searchlight". We try to focus on those key concepts and conventions that are mentioned only in passing, or not given a precise definition, and explain them.

2 Chapters 1–5

2.1 Classical symmetries and Noether's theorem

Symmetry in the classical sense: a classical physical system is given by (M, ω, H), where M is a symplectic manifold, ω the symplectic form, and H the Hamiltonian. A continuous symmetry is a vector field X on M whose infinitesimal flow preserves ω; in the traditional sense one requires $[X, X_H] = 0$. Noether's theorem tells us that there exists $f \in \Gamma(M)$ with $\iota_X \omega = df$ (with $H^1(M)$ as an obstruction; in the field-theoretic context this obstruction disappears, and we will not dwell on it here).

For conformal symmetry we extend the definition slightly: a symmetry is a Lie algebra $\mathfrak{g}$ together with a moment map $\mathfrak{g} \to \Gamma(M)$ that is a Lie algebra homomorphism to the Poisson bracket, and a distinguished element $h \in \mathfrak{g}$ whose image is H. Note that we do not require h to commute with all elements. The purpose of this definition is to include symmetries such as Lorentz boosts.

2.2 Noether's theorem in field theory

This section is an abbreviated account of Section 2.4.2.

Consider a one-parameter symmetry, and assume the symmetry has a "pointwise" form, for example a translation $x \to x + \delta\omega$. We introduce a non-rigid version $x \to x + \delta\omega(x)$. Then for any Φ, when $\delta\omega(x)$ is constant we have $\delta S = 0$; we therefore guess that δS, as a linear functional of $\delta\omega(x)$, can depend only on $\partial_\mu \delta\omega(x)$, i.e. it has the form

$$\delta S = -\int d^d x \, j^\mu(x;\Phi)\, \partial_\mu \delta\omega(x).$$

Integrating by parts,

$$\delta S = \int d^d x \, \partial_\mu j^\mu(x;\Phi)\, \delta\omega(x).$$

When Φ satisfies the equations of motion, $\delta S = 0$, and since $\delta\omega(x)$ is arbitrary, we get $\partial_\mu j^\mu(x;\Phi) = 0$.

2.3 Quantization and quantum symmetries

Given the physical system above, a quantization is a procedure lifting $\Gamma(M)$ to operators on some Hilbert space, satisfying

$$[\hat{f}, \hat{g}] = i\hbar\widehat{\{f, g\}} + O(\hbar^2).$$

To lift a classical symmetry to a quantum one, one first uses Noether's theorem to find f and then lifts it to $\hat{f}$. The commutation relation is then broken (namely by the $O(\hbar^2)$ term), but one may find a weakened condition (a so-called soft breaking):

$$[\hat{f}, \hat{g}] = i\hbar\widehat{\{f, g\}} + c,$$

where c is a scalar.

Remark 1. For the situation we care about, namely symmetries induced by coordinate transformations, we give another interpretation of soft breaking, via the Ward identity:

$$[Q_a, \Phi] = -iG_a\Phi.$$

Since the G_a obey the commutation relations of the classical symmetry, we have $[[Q_a, Q_b], \Phi] = -iG_{[a,b]}\Phi$. Hence $[Q_a, Q_b] = Q_{[a,b]} + c$. For the meaning of the notation see equation (2.163).

2.4 The two-dimensional conformal group and the Witt algebra

We state the conclusion directly: writing $z = x + yi$, the infinitesimal two-dimensional conformal transformations are generated by $z \mapsto z + \epsilon f(z)$, where f is a holomorphic function with expansion

$$f(z) = \sum_{n=-\infty}^{\infty} f_n z^n.$$

For $\bar{z} = x - yi$, we have $\bar{z} \mapsto \bar{z} + \epsilon\bar{f}(\bar{z})$, where $\bar{f}(\bar{z}) = \sum_{n=-\infty}^{\infty} \bar{f}_n \bar{z}^n$.

After complexification, $z \mapsto z + \epsilon f(z)$, $\bar{z} \mapsto \bar{z} + \epsilon g(\bar{z})$, with f and g two independent holomorphic functions.

Remark 2 (On treating $z, \bar{z}$ as independent variables). We give a rigorous model for treating $z, \bar{z}$ as independent variables:

First regard x, y as complex numbers, denoted $x_{\mathbb{C}}, y_{\mathbb{C}}$. For example, a classical field $\phi(x, y)$ is analytically continued to $\mathbb{C}^2$ as $\phi(x_{\mathbb{C}}, y_{\mathbb{C}})$. Then $z = x_{\mathbb{C}} + y_{\mathbb{C}} i$, $\bar{z} = x_{\mathbb{C}} - y_{\mathbb{C}} i$ is simply a one-to-one linear transformation, and after this treatment $z, \bar{z}$ become independent variables.

Thus the two-dimensional conformal group is generated by $z \mapsto z + \epsilon z^n$, $\bar{z} \mapsto \bar{z}$ and $z \mapsto z$, $\bar{z} \mapsto \bar{z} + \epsilon \bar{z}^n$. Denote these by $l_n, \bar{l}_n$. They satisfy the Witt algebra relations:

$$\begin{aligned}
[l_n, l_m] &= (n-m) l_{m+n}, \\
[\bar{l}_n, \bar{l}_m] &= (n-m) \bar{l}_{m+n}, \\
[l_n, \bar{l}_m] &= 0.
\end{aligned}$$

2.5 The Virasoro algebra

Taking the central extension of the Witt algebra gives the Virasoro algebra:

$$\begin{aligned}
[L_n, L_m] &= (n-m) L_{m+n} + \frac{c}{12} m(m^2-1) \delta_{m+n,0}, \\
[\bar{L}_n, \bar{L}_m] &= (n-m) \bar{L}_{m+n} + \frac{c}{12} m(m^2-1) \delta_{m+n,0}, \\
[L_n, \bar{L}_m] &= 0.
\end{aligned}$$

Definition 3. A 2d QFT has conformal symmetry if and only if there exist operators $L_n, \bar{L}_n$ satisfying the above commutation relations, with $L_{-1} = H + iL$, $\bar{L}_{-1} = H - iL$, where H, L are the Hamiltonian and the momentum operator.

2.6 OPE and commutation relations

If one naively uses Taylor expansion, one may write expressions such as

$$\phi(z)\psi(w) = \phi(z)\big(\psi(z) + (z-w)\partial_z \psi(z) + \cdots\big) = \phi(z)\psi(z) + (z-w)\phi(z)\partial_z \psi(z) + \cdots$$

This concerns the problem of expanding the same formal series in different domains; for a rigorous account see Chapter 3 of *Vertex Algebras and Algebraic Curves*. For a concrete example see Section 2.3.1.

The following computation will recur:

Theorem 4 (*Vertex Algebras and Algebraic Curves*, Proposition 3.3.1)**.** *If*

$$\phi(z)\psi(w) = \sum_{j=0}^{N-1} \frac{\gamma_j(w)}{(z-w)^{j+1}} + \text{reg.},$$

then

$$[\phi(z), \psi(w)] = \sum_{j=0}^{N-1} \frac{1}{j!} \gamma_j(w) \partial_w^j (z-w),$$

and vice versa.

If we expand $\phi(z) = \sum_n \phi_n z^{-n}$*, then*

$$[\phi_n, \psi_m] = \sum_k \binom{n}{m+n-k} (\gamma_{m+n})_k.$$

3 Chapter 6

3.1 Radial quantization

3.1.1 Why it is called quantization

As discussed above, quantization is the process of lifting classical observables to operators. By comparison, the difference between radial quantization and time-ordered quantization is as follows: consider the correlation function $\langle \phi_1(z_1) \dots \phi_n(z_n) \rangle$; it is equal under the different quantizations. Under the different quantizations the correlation function is respectively

$$\langle 0 | \, \mathcal{R}\big(\phi_{1R}(z_1) \dots \phi_{nR}(z_n)\big) \, |0\rangle$$

and

$$\langle 0 | \, \mathcal{T}\big(\phi_{1T}(z_1) \dots \phi_{nT}(z_n)\big) \, |0\rangle \, ,$$

where ϕ_{iR} is the operator in radial quantization and ϕ_{iT} the operator in time-ordered quantization; to obtain the correct correlation function one must order the operators in the correct way for each.

Remark 5. The author believes that the Hilbert spaces arising from the two quantizations cannot be compared with each other.

3.1.2 Why radial quantization?

In radial quantization the equal-time surface is bounded, so one can define contour integrals.

4 Chapters 7–8

4.1 Representations of the Virasoro algebra

4.1.1 Prelude: representations of SL_2

4.2 Why is the minimal model unique?*

In Chapter 6 we proved that for a CFT, given c, all the h_i, and the structure constants C_{ijk}, one can compute all correlation functions, and thus determine the CFT. In this chapter we will prove that, based on the fusion rules and associativity, one can solve for all the C_{ijk} (up to normalization/sign).

We start from the familiar picture.

(Figure: CrossSymCFT.png)

Say the linear relation between those graphs with four legs is $F \begin{bmatrix} i & j \\ l & m \end{bmatrix}_{p,q}$; then

$$C_{ij}^p C_{lm}^p = \sum_q F \begin{bmatrix} i & j \\ l & m \end{bmatrix}_{p,q} C_{im}^q C_{jl}^q .$$

The problem we care about is how to solve this overdetermined system, with the normalization condition $C_{ii}^0 = 1$. One could consider the case $i = j = k = l$; here only $C_{ii}^{p\,2}$ is involved, and the equation is linear in these quantities. If for i the fusion number $N_{ii}^0 = 1$—which holds for all i in

the minimal model—then the term C_{ii}^0 appears, so all structure constants such as C_{ii}^p are fixed, under some degeneracy assumption on F.

Next consider $i = j \neq k = l$; then the left-hand side of the equation is known and the right-hand side involves only C_{ij}^q [2], so all structure constants are fixed.

In the thesis "The Structure Constants of the Minimal Models" by Benjamin Gerraty (https://researchers.ms.unimelb.edu.au/~dridout@unimelb/students/Gerraty.MScThesis.pdf) some examples are computed.

5 Chapter 9

5.1 Contour integrals of multivalued functions

The (integrands of the) multivalued functions considered in this chapter have the form $f(w) = w^a(w-1)^b(w-z)^c$, where a, b, c are rational numbers. Such a function can be defined single-valuedly on $\mathbb{C} \setminus L$, where L consists of three lines from $\{0, 1, z\}$ to infinity, the so-called branch cuts. We call this value the principal value; crossing a branch cut changes it by a phase.

For a path $l : [0, 1] \to \mathbb{C}$ avoiding these singularities, we can define a continuous function $f|_l$ on the path as follows: take the principal value at the starting point, and each time the path crosses a branch cut, multiply by the corresponding phase so that the value stays continuous.

For a closed path, a legitimate contour integral requires that the value at the start and end points coincide; for general a, b, c this amounts to the winding numbers of $\{0, 1, z\}$ all being 0.

One can also adopt the Riemann surface point of view, which makes the story simpler: the Riemann surface of f is the universal cover of $\mathbb{C} \setminus \{0, 1, z\}$, whose fundamental group is the free group $G = \langle a, b, c \rangle$ on three generators. Since the monodromy group is abelian, we can quotient the Riemann surface by the derived subgroup of G. A path in $\mathbb{C}$ lifts to the Riemann surface, and a closed path lifts if and only if the deck transformation lies in the derived subgroup, i.e. the winding numbers are 0.

As an example, consider the following contour:

1. without crossing any branch cut, go from near 0 to near 1;
2. cross the branch cut at 1 from above to below, then return from near 1 to near 0;
3. cross the branch cut at 0 from above to below, going from near 0 to near 1;
4. cross the branch cut at 1 from below to above, then return from near 1 to near 0;
5. cross the branch cut at 0 from below to above, and return to the origin.

Let crossing 0 from above to below change the phase by θ_1, and crossing 1 from above to below change the phase by θ_2; then the contour integral equals

$$(1 - e^{i\theta_1} + e^{i(\theta_1+\theta_2)} - e^{i\theta_2}) \int_0^1 f \, dz.$$

6 Chapter 10

First note that modular invariance is an axiom, in fact equivalent to being Weyl-anomaly free. One can construct "legitimate" conformal field theories that are not modular invariant: from the VOA point of view one only needs the fusion product to be closed; the simplest example is the minimal model taking only the Virasoro module $\phi_{(1,1)} \times \bar{\phi}_{(1,1)}$.

The theme of this chapter is to examine what constraints modular invariance imposes on a CFT.